MATH 127 Calculus for the Sciences

Lecture 33

Today's lecture

Last time
Linear approximation error
Taylor's theorem

This time
Course note coverage Section 5.1
Integration by substitution

Information
Quiz 10 on Wednesday.

Differentials

Let us recall some of the concepts that will show up today.
Suppose $y = f(x)$ is a function in the variable x. Then we have

$$dy = \boxed{}.$$

Example

1. If we let $t = \cos(x)$, then $dt = \boxed{}$.
2. If we let $u = \ln v$, then $du = \boxed{}$ and $dv = \boxed{}$.

Domain and range

Let $y = 2x$.

If the domain for x is $[0, 1]$, then the range for y is $\boxed{}$.

$\Updownarrow$ is the same as saying

As x moves from 0 to 1, y moves from $\boxed{}$.

Let $v = -\ln t$. Then

If the domain for t is $[1, e]$, then the range for y is $\boxed{}$.

$\Updownarrow$ is the same as saying

As t moves from 1 to e, v moves from $\boxed{}$.

Integration by substitution

We can sometimes make an integral easier by performing **a change of variable**, also called **Integration by substitution**. This is typically used when there is a composition of functions $f(g(x))$, but is sometimes useful even when there is not.

Algorithm. If the integrand of an integral contains a composition of functions $f(g(x))$, then

1. Let $u = g(x)$ and compute $du = g'(x)\,dx$.
2. Using these to convert the original integral entirely into terms of the new variable u.
   2.1 If you have a definite integral, then you also need **new bounds**: ask yourself as x moves from the old bounds, u moves from where to where?
3. Solve the new integral in terms of u.
4. Substitute back in for x.
   4.1 If you have a definite integral, your result is just a number, so there is no need for substitution.

Example 1

Example Evaluate $\int x^2 \cos(x^3 + 2)\, dx$.

We have a composition of functions: $\cos(x^3 + 2)$.
So we should try to substitute

$$u = \boxed{} \qquad du = \boxed{}$$

Then

$$\int x^2 \cos(x^3 + 2)\, dx = \int x^2 \cos(u)\, dx$$

What is this step for? $= \int \cos(u) \cdot \frac{1}{3} \cdot (3x^2\, dx)$

$$= \boxed{}$$

$$= \boxed{}$$

$$= \boxed{}.$$

Example 1: check

Example Evaluate $\int x^2 \cos(x^3+2)\,dx$. In the previous slide we found the answer $\frac{1}{3}\sin(x^3+2)+C$.

If you are unsure, you can check your antiderivative by differentiating.

$$\begin{aligned}\frac{d}{dx}\left(\frac{1}{3}\sin(x^3+2)+C\right) &= \frac{1}{3}\cos(x^3+2)\cdot 3x^2 \quad \text{(chain rule)}\\ &= x^2\cos(x^3+2),\end{aligned}$$

which is the original integrand.

Exercise 1

What substitution should you use to evaluate each integral?

(a) $\displaystyle\int 2e^x\sqrt{e^x+1}\,dx$

(b) $\displaystyle\int \frac{2x}{1+x^2}\,dx$

(c) $\displaystyle\int \cos^2(x)\,\sin(x)\,dx$

Exercise 2(c): solution

Example Evaluate $\int \cos^2(x)\, \sin(x)\, dx$

Substitute

$$u = \cos(x) \qquad du = \boxed{}$$

Then,

$$\begin{aligned}\int \cos^2(x)\, \sin(x)\, dx &= \int u^2 \sin(x) dx \\ &= \int \boxed{} \\ &= \boxed{} \\ &= \boxed{}\end{aligned}$$

Example 2: Definite integral

Example Calculate the area under $y = \dfrac{4x}{1+x^2}$ from $x = 0$ to $x = 1$.

The area is just given by a definite integral

$$\boxed{}$$

Substitute

$$u = 1 + x^2 \qquad du = \boxed{}$$

But before we do our substitution, remember, for definite integrals, we also have to change the bounds.

1. When $x = 0$, $u = 1$.
2. When $x = 1$, $u = 2$.

So as x moves from 0 to 1, u moves from $\boxed{}$.

Example 2: (cont.)

Now let's substitute

$$u = 1 + x^2 \qquad du = 2x\,dx$$

As x moves from 0 to 1, u moves from $\boxed{}$.

$$\int_0^1 \frac{2x}{1+x^2}\,dx = \int_0^1 \frac{1}{1+x^2} \cdot 2x\,dx$$

$$= \int_{\boxed{}}^{\boxed{}} \frac{1}{u}\,du.$$

Remark This you know how to solve. Again note that the result will be a number, so unlike before, we do not have to do a reverse substitution in the end.

Example 3

Example Evaluate $\displaystyle\int_1^{e^{\pi/4}} \frac{\sec^2(\ln(x))}{x}\,dx$.

Make substitution

$$u = \boxed{} \qquad du = \boxed{}$$

As x moves from 1 to $e^{\pi/4}$, u moves from $\boxed{}$

$$\begin{aligned}\int_1^{e^{\pi/4}} \frac{\sec^2(\ln(x))}{x}\,dx &= \int_1^{e^{\pi/4}} \sec^2(\ln(x))\,\frac{1}{x}\,dx \\ &= \int_{\boxed{}}^{\boxed{}} \sec^2(u)\,du \\ &= \boxed{} \\ &= \boxed{}\end{aligned}$$

Summary

Here are the ways you can solve integrals

1. Guess and check
2. Known trig formulas that you have to memorize
3. Integration by substitution, don't forget to substitute back the original variable in the end.

For definite integrals, you can try

1. Everything above
2. For integral by substitution, don't forget to change the bound.